THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 10: **The Human Eye and the Colourful World** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If the earth had no atmosphere, the duration of the day would be 1

(a) more **(b)** slightly less **(c)** double **(d)** same

2. Assertion (A): The sky appears dark to an astronaut. Reason (R): There is no atmosphere in outer space to scatter sunlight. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

3. The Moon seen from the earth does not twinkle because 1

(a) it is a planet **(b)** it is much closer and extended
(c) it has no atmosphere **(d)** it emits its own light

4. A myopic eye gets a lens of -2 D. The correct statement is 1

(a) the far point is infinity **(b)** the near point is 50 cm
(c) the lens is convex **(d)** the far point of the eye is 50 cm

5. Assertion (A): A bifocal lens has both concave and convex parts. Reason (R): A person may need correction for both distant and near vision. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

6. A person with presbyopia and myopia uses 1

(a) plane glasses **(b)** a single convex lens only
(c) a single concave lens only **(d)** bifocal lenses

7. For a person who uses +2.5 D reading glasses (reading at 25 cm), the near point of unaided eye is 1

(a) 25 cm **(b)** 100 cm **(c)** about 67 cm **(d)** 40 cm

8. Assertion (A): A rainbow is seen when the Sun is behind the observer. Reason (R): Raindrops disperse sunlight and return the coloured light back towards the side from which the sunlight came. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

9. **Assertion (A):** A person with myopia cannot see distant objects clearly. 1
Reason (R): In myopia the image of a distant object is formed in front of the retina.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** Danger signal lights are red in colour. 1
Reason (R): Red light is scattered the most by fog and smoke.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A lens of power -3 D corrects a myopic eye. Find the far point without the lens. 2

12. Why is violet light deviated more than red light by a prism? 2

13. In an experiment a beam of red light is passed through a prism and no spectrum forms. Give a reason. 2

14. Why does a ray of white light not form a spectrum in a glass slab but forms one in a prism? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain with a diagram description the path of light in a water droplet that forms a primary rainbow. 3

16. Why do you see the sky in a clear and dust-free area darker blue than in a dusty place? 3

17. Explain how a person with myopia may test whether a lens is concave or convex by looking through it. 3

OR

A person's eye can form a clear image on the retina if the eye lens focal length is 2.0 cm, for a distant object. For an object at 25 cm, what should be the focal length of the lens (retina distance is 2.0 cm)? Is this possible if the minimum is 1.85 cm? 3

18. How can the blurred vision in a person with cataract be cured, and why can't spectacles correct it? 3

OR

Which cells of the retina work in dim light and which in bright light? What do they do? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** At Jaisalmer, Arjun watched the Sun set over the desert and found that the Sun looked red near the horizon but white at noon. The sky overhead looked blue. From the same place, astronauts in space see the sky as dark. 5
(i) Name the phenomenon responsible for the blue colour of the sky.
(ii) Why do we see the sky blue and not red?
(iii) Why is red light used for danger signals?
(iv) Explain why the Sun appears red at sunrise and sunset and why the sky appears dark to astronauts.

20. Explain why the setting Sun appears red and flattened. Also explain why we see the Sun even after it has set below the horizon. 5

OR

A person with myopia has a far point at 125 cm and a person with hypermetropia has a near point at 100 cm. For each find the focal length and power of the corrective lens (reading distance 25 cm for the second) and compare the nature of the lenses. 5

END OF QUESTION PAPER — Total: 40 Marks